

SIDNEY ELUWA PROJECT

# Customer RFM Segmentation & BI Dashboard

Google BigQuery • SQL • Power BI

2025 Full-Year Sales Data | 285 Customers | 994 Orders | £16,694 Revenue

| Tools Used                | Dataset                    | Time Period    | Deliverable                 |
|---------------------------|----------------------------|----------------|-----------------------------|
| BigQuery • SQL • Power BI | 994 orders • 285 customers | Jan – Dec 2025 | BI Dashboard + SQL Pipeline |

## 1. Project Overview

This project delivers an end-to-end customer segmentation pipeline and interactive BI dashboard for an online print and personalised-products retailer. Using twelve months of transactional data (January–December 2025), the project applies an RFM (Recency, Frequency, Monetary) framework to classify 287 customers into eight clearly defined behavioural segments, enabling the marketing team to take targeted, data-driven action on each group.

|          |                                                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem  | The business had no systematic way to differentiate its customer base. All customers received identical marketing communications regardless of their purchase history, resulting in wasted budget on inactive customers and under-investment in high-value ones. |
| Solution | Design and deploy an automated SQL pipeline in Google BigQuery that ingests monthly sales files, computes RFM scores, assigns decile-based rankings, and writes a production-ready segmentation table — surfaced in a Power BI dashboard for ongoing monitoring. |
| Outcome  | A repeatable, low-maintenance pipeline that classifies every customer into one of eight segments (VIPs through Inactive) and a live Power BI report stakeholders can filter by segment.                                                                          |

## 2. Business Context & Objectives

### 2.1 Background

The retailer sells six product lines — Flyers, Canvas Prints, Greeting Cards, Business Cards, Posters, and Photo Books — entirely through an e-commerce channel. With 994 orders placed across 2025 and

an average order value of £17.17, the business relies on repeat purchasing to drive profitability. Without visibility into which customers are at risk of churning or which are ready for an upsell, the marketing team was spending budget inefficiently.

## 2.2 Business Objectives

- Identify and protect the top-tier customers (VIPs, Champions) who drive the majority of revenue.
- Flag at-risk customers (Attention, Risky) early enough for a re-engagement campaign to succeed.
- Stop wasting budget on Inactive customers and re-allocate it to high-potential Promising segments.
- Give the marketing team a self-service dashboard they can refresh without analyst involvement.

## 2.3 Success Criteria

- A fully automated BigQuery pipeline that produces a clean segmentation table with zero manual steps after initial deployment.
- A Power BI dashboard that loads within 5 seconds and allows segment-level drill-through.
- Clear documentation allowing another analyst to maintain or extend the pipeline.

# 3. Dataset & Data Sources

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## 3.1 Source Files

The input data comprised twelve separate monthly CSV exports (202501.csv through 202512.csv) produced by the transactional system. Each file represented one calendar month of order activity and was uploaded into Google BigQuery as an individual table (sales202501 through sales202512).

| Metric           | Value               | Description                                             |
|------------------|---------------------|---------------------------------------------------------|
| Schema           | 5 columns           | OrderID, CustomerID, OrderDate, ProductType, OrderValue |
| Total orders     | 994                 | Across all 12 monthly files after deduplication         |
| Unique customers | 285                 | Distinct CustomerIDs observed across the full year      |
| Total revenue    | £16,694.11          | Sum of OrderValue across all valid rows                 |
| Avg order value  | £17.17              | Mean transaction size across all 994 orders             |
| Order range      | £4.01 – £48.00      | Min/Max individual order values                         |
| Date range       | Jan 1 – Dec 31 2025 | Full calendar year, analysis date set to 24 Mar 2026    |

## 3.2 Product Mix

Six product types were present in broadly even proportions, confirming no single product category was artificially dominant and that RFM scores would reflect genuine customer behaviour rather than product-driven spikes:

- Greeting Card — 188 orders (19%)
- Business Card — 178 orders (18%)
- Flyer — 163 orders (16%)
- Poster — 159 orders (16%)
- Canvas Print — 155 orders (16%)
- Photo Book — 151 orders (15%)

### 3.3 Data Quality Issues

One data quality issue was identified during ingestion: the December 2025 file (202512.csv) contained four trailing empty columns beyond the standard five-column schema. This was handled in the BigQuery by dropping the empty columns, so there will be a consistent schema in the pipeline.

```
-- December file: trailing empty columns were dropped
ALTER TABLE `rfm-2026-01.sales.sales202512`
DROP COLUMN string_field_5;

ALTER TABLE `rfm-2026-01.sales.sales202512`
DROP COLUMN string_field_6;

ALTER TABLE `rfm-2026-01.sales.sales202512`
DROP COLUMN string_field_7;

-- All months having: SELECT * (schema consistent)
SELECT * FROM `rfm-2026-01.sales.sales202501`
```

## 4. Methodology

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### 4.1 Why RFM?

RFM is a proven, interpretable segmentation framework widely used in e-commerce and direct marketing. It decomposes customer value into three orthogonal dimensions:

| Metric               | Value                 | Description                                                                  |
|----------------------|-----------------------|------------------------------------------------------------------------------|
| <b>Recency (R)</b>   | Days since last order | Lower is better — a customer who bought recently is more likely to buy again |
| <b>Frequency (F)</b> | Total order count     | Higher is better — more transactions signal habit and loyalty                |
| <b>Monetary (M)</b>  | Total spend (GBP)     | Higher is better — captures the financial value of the relationship          |

RFM was chosen over more complex models (e.g. CLV prediction, clustering) because: (a) it requires no machine learning infrastructure, (b) it produces human-readable, actionable segments, and (c) it can be maintained and extended by any SQL-literate analyst.

### 4.2 Scoring Approach — Deciles

Each customer was ranked across the full 287-customer base on each RFM dimension and then assigned a decile score from 1 (worst) to 10 (best) using BigQuery's NTILE(10) window function. This relative scoring approach — rather than fixed thresholds — ensures the score distribution adapts automatically as the customer base grows or purchase behaviour shifts month to month.

The three decile scores were summed to produce a total score ranging from 3 to 30. Eight named segments were defined using total score thresholds:

| Segment          | R Score | F Score | M Score | Description & Strategy                                                                |
|------------------|---------|---------|---------|---------------------------------------------------------------------------------------|
| <b>VIPs</b>      | 9-10    | 9-10    | 9-10    | Total score $\geq 28$ . Best customers on all three dimensions. Protect at all costs. |
| <b>Champions</b> | 8-10    | 7-9     | 7-9     | Total score $\geq 24$ . Highly engaged, high spend. Reward and nurture.               |
| <b>Loyalists</b> | 6-9     | 6-9     | 6-9     | Total score $\geq 20$ . Consistent repeat buyers. Upsell and cross-sell.              |
| <b>Promising</b> | 5-8     | 4-7     | 4-7     | Total score $\geq 16$ . Growing relationship. Encourage second purchase cycle.        |
| <b>Active</b>    | 4-7     | 3-6     | 3-6     | Total score $\geq 12$ . Moderate engagement. Monitor and incentivise.                 |
| <b>Attention</b> | 3-5     | 2-5     | 2-5     | Total score $\geq 8$ . Recency slipping. Early re-engagement campaign needed.         |
| <b>Risky</b>     | 2-4     | 2-4     | 2-4     | Total score $\geq 4$ . Significant lapse. Discount/win-back offer required.           |
| <b>Inactive</b>  | 1-3     | 1-2     | 1-2     | Total score $< 4$ . Likely churned. Suppress from main campaigns.                     |

## 5. Technical Implementation

### 5.1 Architecture Overview

The pipeline was designed as a five-step BigQuery workflow progressing from raw ingested CSVs to a BI-ready segmentation table, with intermediate results materialised as Views (no storage cost, always fresh) and the final output as a Table (fast query for Power BI Direct Query).

| # | Step                       | Detail                                                                                                                                                                                        |
|---|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Data Consolidation</b>  | UNION ALL across 12 monthly tables into a single sales_2025 persistent table. December file handled dropping empty columns to ensure schema match.                                            |
| 2 | <b>RFM Metrics + Ranks</b> | CTE computes last_order_date, recency (days), frequency (count), monetary (sum) per CustomerID. ROW_NUMBER() window functions produce r_rank, f_rank, m_rank. Stored as a View (rfm_metrics). |
| 3 | <b>Decile Scoring</b>      | NTILE(10) over the three rank columns assigns r_score, f_score, m_score (1=worst, 10=best). Stored as View rfm_scores.                                                                        |

| # | Step                      | Detail                                                                                                                                                                                             |
|---|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <b>Total Score</b>        | r_score + f_score + m_score summed into total_score (range: 3-30). Stored as View rfm_total_scores.                                                                                                |
| 5 | <b>Segment Assignment</b> | CASE statement maps total_score ranges to named segments (VIPs, Champions, Loyalists, Promising, Active, Attention, Risky, Inactive). Written to persistent Table rfm_segments_final for Power BI. |

## 5.2 Key SQL Patterns

### Step 2 — CTE + Window Functions for RFM Metrics

```
CREATE OR REPLACE VIEW rfm-2026-01.sales.rfm_metrics AS
WITH current_date AS (
    SELECT DATE('2026-03-24') AS analysis_date
),
rfm AS (
    SELECT
        CustomerID,
        MAX(OrderByDate) AS last_order_date,
        DATE_DIFF((SELECT analysis_date FROM current_date),
            MAX(OrderByDate), DAY) AS recency,
        COUNT(*) AS frequency,
        SUM(OrderByValue) AS monetary
    FROM rfm-2026-01.sales.sales_2025
    GROUP BY CustomerID
)
SELECT
    rfm.*,
    ROW_NUMBER() OVER(ORDER BY recency ASC) AS r_rank,
    ROW_NUMBER() OVER(ORDER BY frequency DESC) AS f_rank,
    ROW_NUMBER() OVER(ORDER BY monetary DESC) AS m_rank
FROM rfm;
```

#### Design decisions in this step:

- CTE for analysis\_date — using a subquery CTE for the reference date means the analysis\_date can be updated in a single place when the pipeline is re-run in future months.
- ROW\_NUMBER() over RANK() — avoids ties inflating scores; every customer gets a unique ordinal position.
- Recency sorted ASC — lower recency (more recent) = lower rank number; the subsequent NTILE reversal (ORDER BY r\_rank DESC) then correctly assigns 10 to the most recent customers.

### Step 3 — Decile Scoring with NTILE

```
CREATE OR REPLACE VIEW rfm-2026-01.sales.rfm_scores AS
SELECT *,
    NTILE(10) OVER (ORDER BY r_rank DESC) AS r_score,
    NTILE(10) OVER (ORDER BY f_rank DESC) AS f_score,
    NTILE(10) OVER (ORDER BY m_rank DESC) AS m_score
FROM `rfm-2026-01.sales.rfm_metrics`;
```

The rank-then-score approach (ROW\_NUMBER in Step 2, NTILE in Step 3) is a deliberate two-step pattern. Applying NTILE directly to the raw metric values can produce skewed deciles when the distribution is non-normal (e.g. a small number of very high-spend customers). Ranking first normalises the distribution before NTILE divides it into equal buckets.

## Step 5 — Segment Assignment

```
CASE
  WHEN total_score >= 28 THEN 'VIPs'
  WHEN total_score >= 24 THEN 'Champions'
  WHEN total_score >= 20 THEN 'Loyalists'
  WHEN total_score >= 16 THEN 'Promising'
  WHEN total_score >= 12 THEN 'Active'
  WHEN total_score >= 8  THEN 'Attention'
  WHEN total_score >= 4  THEN 'Risky'
  ELSE                      'Inactive'
END AS rfm_segment
```

## 5.3 View vs Table — Storage Strategy

Intermediate steps (Steps 2–4) are materialised as Views rather than Tables. This means BigQuery always re-computes them from the source data when queried — there is no stale data risk and no additional storage cost. The final output (Step 5) is a Table, providing fast query performance for the Power BI connection without re-running the full chain on every dashboard load.

# 6. Power BI Dashboard

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## 6.1 Dashboard Design

The Power BI report connects directly to the rfm\_segments\_final table in BigQuery and is structured around three primary analytical questions:

- Segment distribution — How many customers sit in each RFM segment and what proportion of total revenue do they represent?
- Individual customer drill-through — Which specific customers are in a given segment and what are their RFM scores?

## 6.2 Key Visuals

- Segment summary KPI cards— Cards to reveal segment customer count and revenue contribution.
- Bar chart — Visual breakdown of customer count by segment to highlight any segment where headcount and revenue proportions diverge (e.g. few VIP customers representing outsized revenue).
- Customer-level table — Sortable, filterable table of all CustomerIDs with their recency, frequency, monetary, and segment values for operational use (e.g. generating a campaign export list).

## 7. Challenges & Solutions

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| # | Step                                    | Detail                                                                                                                                                                                                                                                                     |
|---|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Schema inconsistency in Dec file</b> | The December CSV contained 4 trailing empty columns. Solved by dropping the columns to conform to the schema of the other 11 months.                                                                                                                                       |
| 2 | <b>NTILE on skewed distributions</b>    | Direct NTILE on raw monetary values would cluster most customers in lower deciles due to a long tail of high spenders. Solved by the two-step rank-then-score pattern: ROW_NUMBER() first normalises the distribution, then NTILE divides it evenly.                       |
| 3 | <b>Analysis date management</b>         | Hardcoding TODAY() in a production pipeline causes different outputs each day. Solved using a CTE that externalises the analysis_date as a single, clearly documented parameter — easy to update and review in version control.                                            |
| 4 | <b>View chain performance</b>           | The nested View chain (rfm_metrics -> rfm_scores -> rfm_total_scores) re-computes fully on each query. Solved by materialising only the final output as a Table, keeping intermediate Views for cost-free freshness and materialising the Power BI-facing layer for speed. |
| 5 | <b>Segment threshold calibration</b>    | Initial thresholds produced unbalanced segments. Iteratively adjusted total_score boundaries by reviewing the distribution of scores in rfm_total_scores and aligning to roughly equal-sized strategic groups.                                                             |

## 8. Key Results & Business Insights

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### 8.1 Pipeline Output

The final rfm\_segments\_final table contains one row per customer (285 rows), with CustomerID, recency, frequency, monetary, r\_score, f\_score, m\_score, total\_score, and rfm\_segment columns — everything required for campaign execution and BI reporting without any further transformation.

### 8.2 Indicative Findings

- Revenue concentration: a relatively small number of VIP and Champion customers are expected to account for a disproportionate share of the £16,694 total revenue — validating the business case for differentiated treatment.
- Re-engagement opportunity: the Attention and Risky segments, identified by declining recency despite historical frequency, represent a quantified pool of customers who may respond to a win-back campaign (discount code, personalised communication).
- Low-risk suppression: the Inactive segment can be excluded from paid campaigns immediately, reducing wasted spend without risk of losing engaged customers.
- Product affinity: the even spread across six product types means targeted product recommendations (e.g. promoting Photo Books to Canvas Print buyers) could increase cross-sell revenue.

## 8.3 Recommended Next Steps

1. Deploy campaign actions for VIP/Champions: exclusive loyalty reward or early access offer.
2. Launch re-engagement sequence for Attention/Risky: time-limited discount with a 30-day response window.
3. Suppress Inactive from all paid channels: reallocate budget to Promising segment acquisition.
4. Schedule monthly pipeline refresh: update analysis\_date CTE, re-run Steps 1-5, and refresh Power BI.
5. Extend to CLV modelling: use RFM scores as features in a Customer Lifetime Value regression to prioritise acquisition lookalike audiences.

## 9. Skills Demonstrated

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| Metric                    | Value        | Description                                                                                           |
|---------------------------|--------------|-------------------------------------------------------------------------------------------------------|
| SQL (BigQuery)            | Advanced     | CTEs, window functions (ROW_NUMBER, NTILE), UNION ALL, Views vs Tables, DDL (CREATE OR REPLACE)       |
| Data Modelling            | Intermediate | Star-schema awareness, separation of raw/transformed/serving layers, View chain design                |
| Business Analysis         | Intermediate | Requirements gathering, RFM framework knowledge, segment threshold design, actionable insight framing |
| Power BI                  | Intermediate | BigQuery connector, data model, report design, filter, DAX measures for segment KPIs                  |
| Data Quality Management   | Intermediate | Schema mismatch detection and targeted resolution without pipeline-wide refactoring                   |
| Documentation             | Strong       | Inline SQL comments, step-by-step pipeline documentation, self-describing View naming convention      |
| Stakeholder Communication | Strong       | Translating analytical output into business-language segment definitions and recommended actions      |



## 10. Project Files

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| Metric                     | Value       | Description                                                                      |
|----------------------------|-------------|----------------------------------------------------------------------------------|
| Big_Query_RFM_workflow.sql | BigQuery    | Complete 5-step SQL pipeline from raw ingestion to final segmentation table      |
| RFM-analysis.pbix          | Power BI    | Dashboard connected to rfm_segments_final with all report pages and DAX measures |
| 202501.csv – 202512.csv    | Source data | 12 monthly transaction files used as pipeline input (anonymised CustomerIDs)     |

**Reproducibility:** to re-run the pipeline, update the analysis\_date value in the Step 2 CTE to the desired reference date, ensure all 12 monthly tables are loaded into the rfm-2026-01.sales BigQuery dataset, and execute the script sequentially. No additional configuration is required.